

Library API - REST and GraphQL Cheat sheet

Žan Perat

1 Library REST API

REST examples (use -k with self-signed cert)

```
# List authors (public)
curl -k https://192.168.11.104:3443/authors

# Get author by id
curl -k https://192.168.11.104:3443/authors/1

# Get books for author
curl -k https://192.168.11.104:3443/authors/1/books

# Create author (requires librarian/admin)
curl -i -X POST http://192.168.11.104:3000/authors \
  -H "Content-Type: application/json" \
  -u username \
  -d '{"name":"New Author","birth_year":1970,"nationality":"Country"}'

# Update author
curl -i -X PUT http://192.168.11.104:3000/authors/2 \
  -H "Content-Type: application/json" \
  -u username \
  -d '{"name":"Updated Name"}'

# Delete author (admin only)
curl -i -X DELETE http://192.168.11.104:3000/authors/5 -u username

# List books (optional genre filter)
curl -k "https://192.168.11.104:3443/books?genre=Fantasy"

# Get book by id
curl -k https://192.168.11.104:3443/books/1

# Create book (requires librarian/admin)
curl -k -u username -H "Content-Type: application/json" \
  -d '{"title":"New Book","author_id":1,"year":2020, \
  "genre":"Sci-Fi","isbn":"978-0-452-28423-4","available":true}' \
  https://192.168.11.104:3443/books
```

```
# Update book
curl -k -u username -X PUT -H "Content-Type: application/json" \
  -d '{"title":"Updated"}' https://192.168.11.104:3443/books/3

# Toggle availability (PATCH)
curl -k -i -X PATCH https://192.168.11.104:3443/books/3/availability \
  -H "Content-Type: application/json" \
  -u username -d '{"available":false}'

# Delete book (admin only)
curl -k -u username -X DELETE https://192.168.11.104:3443/books/3
```

2 Library GraphQL API

GraphQL examples

```
# Public query (POST)
curl -k -X POST -H "Content-Type: application/json" \
  --data '{"query":"{ authors { id name birth_year } }"}' \
  https://192.168.11.104:30443/graphql

# Authenticated mutation (HTTP Basic)
curl -k -u username -X POST -H "Content-Type: application/json" \
  --data '{"query":"mutation { createAuthor(name:\\"New\\") { id name } }"}' \
  https://192.168.11.104:30443/graphql

# GraphiQL IDE (browser)
Open a browser to: https://192.168.11.104:30443/graphql
```

```
#En primer:
{
  authors {
    id
    name
    birth_year
    nationality
  }
}
```